

HDFS (Storage)

A. Hadoop Storage File system

- I. Write a command to create the directory structure in HDFS.
- II. Write a Command to move files from the local Unix/Linux machine to HDFS.

B. Viewing Data Contents, Files and Directory

- a) Write an HDFS command to see the contents of files in HDFS.
- b) Write an HDFS command to see the contents of files which are present in HDFS.

C. Getting Files data from the HDFS to Local Disk

- I. Write an HDFS command to copy the file from HDFS to the local file system.

MapReduce

To Develop a MapReduce program to calculate the frequency of a given word in a given file

Set of data

Bus, Car, bus, car, train, car, bus, car, train, bus, TRAIN,BUS, buS, caR, CAR, car, BUS, TRAIN

PIG

For the given Student dataset perform Relational operations like LOAD, STORE, DUMP, and DESCRIBE Operations in Hadoop Pig framework using Cloudera

StudentID	FirstName	Age	City	CGPA
001	Jagruthi	21	Hyderabad	9.1
002	Praneeth	22	Chennai	8.6
003	Sujith	22	Mumbai	7.8
004	Sreeja	21	Bengaluru	9.2
005	Mahesh	24	Hyderabad	8.8
006	Rohit	22	Chennai	7.8

Perform sales prediction using the ARIMA model.

Data Processing Tool – Hive (NOSQL query-based language)

We have a table 'user data' that contains the following fields:

data date: string

user_id: string

properties: string

The properties field is formatted as a series of attribute=value pairs.

Ex: Age=25; state=Telangana; gender=M;

- i. Create the table in HIVE using hive no sql based query.
- ii. Fill the table with sample data by using some sample databases.
- iii. Write a program that produces a list of properties with minimum value(min_value), largest value(max_value) and number of unique values. Before you start, execute the prepare step to load the data into HDFS.
- iv. Generate a count per state.

Write a program that lists the states and their count from the data input.

SQOOP

It is used to import and export data from SQL to HDFS and Vice versa. Before processing any data first need to import data into HDFS. As of now seen importing files into HDFS but using sqoop tool we can import SQL table data into hive table.

- I. Create table in HIVE using hive query language.**
- II. Import the sql table data into hive using SQOOP tool.**
- III. Export hive table data into local machine and into SQL.**